An exact residual case under an alternating tail

Human Compression project: bounded L044 proof component

Retain the hypotheses and equality parameters of the numerical intransitive-side support theorem. Thus $G \le A_n$ is a transitive connected string C-group, all maximal omissions are intransitive, and a unique eligible interior i-edge ab joins omission orbits A, B. Here $i = k-1 \ge 1$, $h = i+1 = k$, and $A = U \sqcup V$ consists of two faithful natural S_k-orbits of $L = G_{<i}$, matched by $\tau = \rho_{i+1}$. The group L fixes B pointwise. The tail $K = G_{>h}$ fixes exactly $A \cup \{b\}$. Assume additionally that K is the full natural alternating group on $X = B \setminus \{b\}$. Write $U = \{u_1, \dots, u_k\}$, $V = \{v_1, \dots, v_k\}$, with $a = u_k$ and τ exchanging u_j, v_j for every j. Put $\lambda = \prod_j (u_j\ v_j)$ and $s = \rho_i$.

Lemma 1 (The crossing under the diagonal omission)**.** *For $k \ge 4$, evenness forces k even and*

$$s = (u_k\ b) \prod_{j<k} (u_j\ v_j). \tag{1}$$

The case $k = 3$ is impossible. For $k = 2$, formula (1) also holds.

Proof. The tail is nontrivial and generated by involutions, so $q = |X| \ge 5$: the involutions of A_4 generate only its Klein four subgroup, and the smaller alternating groups do not supply a nontrivial involutory generating family. The generator s commutes with K, hence preserves its sole nontrivial orbit X and fixes it pointwise. The natural alternating action has trivial centralizer. Thus $s = (u_k\ b)w$, with w supported on $A \setminus \{u_k\}$.

Let $H = G_{<i-1}$. Its $k-2$ generators fix u_k and v_k. Their faithful images in S_{k-1} are independent, so Whiston's equality clause identifies H as diagonal S_{k-1} on the remaining two rows. For $k \ge 3$ these are the only two fixed points of H in A. String commutation makes w centralize H; since it fixes u_k, it also fixes v_k.

For $k \ge 4$, the centralizer on the remaining rows is C_2, generated by their full exchange. Indeed a commuting permutation permutes the two natural S_{k-1}-orbits, and the natural action has trivial centralizer. Evenness of s forces w to be the exchange, and requires its $k-1$ transpositions to be odd. This proves (1) and the evenness of k.

For $k = 3$, label the two distinguished L generators $\rho_0 = (u_1\ u_2)(v_1\ v_2)$ and $\rho_1 = (u_2\ u_3)(v_2\ v_3)$, as their faithful S_3 action and the fixed anchor permit. An odd involution w on the other four points centralizing ρ_0 must be one of $(u_1\ u_2)$ and $(v_1\ v_2)$. The group G_0 contains $G_{>2}$, which is transitive on B and exchanges every pair $\{u_j, v_j\}$, as well as ρ_1 and s. The permutation ρ_1 joins pairs 2 and 3, w joins pair 1 to pair 2, and the cross-edge of s joins them to B. Hence G_0 is transitive, a contradiction. Finally, for $k = 2$, the only possible odd involution on the three points $A \setminus \{u_2\}$ is a transposition, but w need not initially fix v_2. If it exchanges v_2 with one of u_1, v_1, then $G_0 = \langle s, G_{>1} \rangle$ is transitive: the latter factor exchanges each vertical pair and is transitive on B, while s joins the two pairs and B. Thus intransitivity forces $w = (u_1\ v_1)$, proving (1) in this case too. □

Lemma 2 (A vertical vector from a five-cycle)**.** *Assume (1). If $g = (u_k\ u_j)(v_k\ v_j) \in L$ for $j \ne k$, then*

$$(sg)^5 = \prod_{l \notin \{j,k\}} (u_l\ v_l). \tag{2}$$

For every even $k \ge 4$, the group $P = \langle L, s \rangle$ therefore contains λ.

Proof. On the five points u_k, v_k, u_j, v_j, b, the product sg is a five-cycle. On each other vertical pair it is a transposition. These supports are disjoint, proving (2). Conjugating by diagonal S_k gives the product of all vertical flips except any specified two. Multiplying the products which omit $\{a, c\}$ and $\{b, c\}$ gives the flip of just pairs a, b, for distinct a, b, c. All two-pair flips therefore belong to P. For even k their products include the flip of all pairs, namely λ. □

Proposition 3 (Reduction to a dihedral prefix)**.** *Under the stated equality and alternating-tail hypotheses, necessarily $k = 2$, $i = 1$, $h = 2$, and $G_{<2}$ is exactly the degree-five dihedral marking*

$$\rho_0 = (u_1\ u_2)(v_1\ v_2), \qquad \rho_1 = (u_2\ b)(u_1\ v_1), \qquad |G_{<2}| = 10.$$

This conclusion does not assert that this residual case is realizable with the full string intersection property.

Proof. By Lemma 1, only even k remain. On A, the group $R = G_{>i} = \langle \tau, K \rangle$ acts through $\langle \lambda \rangle$. Since k is even, $\tau|_B$ is even; it moves b into X. The groups $K = A_X$ and $K^\tau = A_{B \setminus \{b^\tau\}}$ generate A_B: their natural alternating supports overlap in at least four points, and the elementary three-cycle gluing argument gives the alternating group on their union. Thus R is a subgroup of $\langle \lambda \rangle \times A_B$ projecting onto both factors. If it did not contain $(\lambda, 1)$, it would be the graph of a nontrivial homomorphism $A_B \to C_2$. Perfection of A_B excludes that possibility, so $\lambda \in R$.

For $k \geq 4$, Lemma 2 also gives $\lambda \in P$. But $P = G_{\leq i}$ and $R = G_{>i}$ have disjoint index sets; full IP requires $P \cap R = 1$. This contradiction excludes every such k. At $k = 2$, the two displayed involutions have product a five-cycle, so generate precisely D_{10} on the five displayed points. They fix X pointwise, proving the stated exact prefix and nothing about its possible completion. □

Source boundary and attribution. This supplies an explicit group-theoretic reduction for the intransitive-side equality step used in L044 Proposition 5.26, physical page 34: Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. It reuses the accepted numerical equality, unique-crossing and Whiston inputs. The vertical-vector word and the disjoint-parabolic IP obstruction are proved here. The surviving D_{10} branch remains a principal obligation; it is not silently discarded by the source's assertion about vertical edges. This note neither proves nor disproves the complete Proposition 5.26 or the main alternating-group rank theorem. The transitive-side equality and all later source branches remain separate. No finite computation, new formal declaration, whole-original or question credit is asserted.